St.Philomena's college
B.Sc V SEM
Unit: 1 MOLECULAR BIOLOGY
Section 1.4
GENETIC CODE

- The three nucleotide (triplet) base sequences in mRNA that act as code words for amino acids in protein constitute the genetic code or simply codons.
- The genetic code is regarded as a dictionary of nucleotide bases (A, G, C and U) that determines the sequence of amino acids in proteins.
- The codons consist of the four nucleotide bases,
  - The purines—adenine (A) and guanine (G), and
  - The pyrimidines—cytosine (C) and uracil (U).
- These four bases produce 64 different combinations

First base 5′end	*Second base (middle one)*				*Third base 3′end*
	U	C	A	G	
U	UUU Phe	UCU Ser	UAU Try	UGU Cys	U
	UUC Phe	UCC Ser	UAC Try	UGC Cys	C
	UUA Leu	UCA Ser	UAA Stop	UGA Stop	A
	UUG Leu	UCG Ser	UAG Stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
	AUG* Met	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G

**AUG serves as initiating codon, besides coding for methionine residue in protein synthesis; UAA, UAG and UGA called as nonsense codons, are responsible for termination of protein synthesis.*

- The nucleotide sequence of the codon on mRNA is written from the 5′- end to 3′ end.
- Sixty one codons code for the 20 amino acids found in protein.

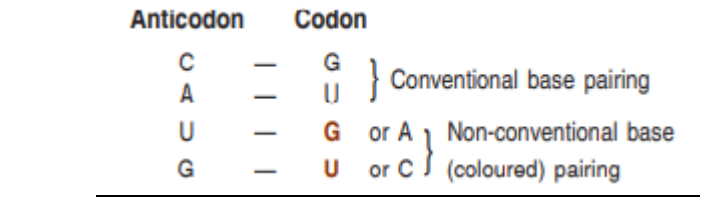
- The three codons UAA, UAG and UGA do not code for amino acids.
    - They act as stop signals in protein synthesis.
    - These three codons are collectively known as termination codons or non-sense codons. The codons UAG, UAA and UGA are often referred to, respectively, *as amber, ochre and opal* codons.
- The codons AUG—and, sometimes, GUG— are the chain initiating codons.

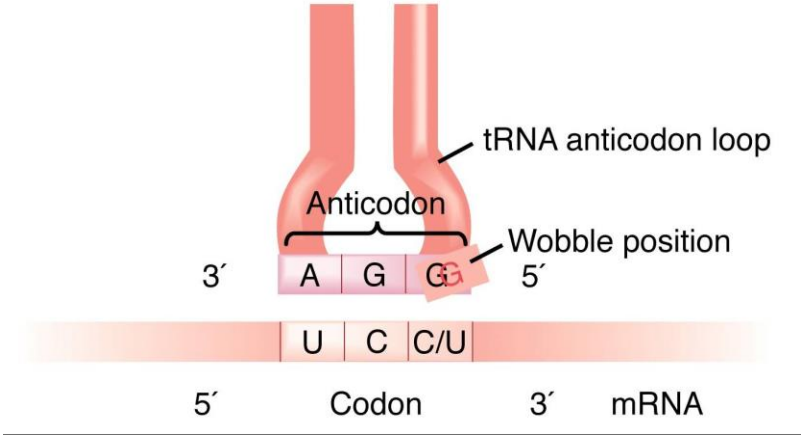
Salient features of the genetic code:

1. The genetic code is a **triplet code** called a codon.
    - The three nucleotide (triplet) base sequences in mRNA that act as code words for amino acids in protein constitute the genetic code or simply codons
2. **Specificity**: A particular codon always codes for the same amino acid, hence the genetic code is highly specific or **unambiguous** e.g. UGG is the codon for tryptophan.
3. **Non-overlapping**:
    - The genetic code is read from a fixed point as a continuous base sequence. It is **non-overlapping, commaless and without any punctuations**.
    - For instance, UUUCUUAGAGGG is read as UUU/CUU/AGA/GGG. Addition or deletion of one or two bases will radically change the message sequence in mRNA. And the protein synthesized from such mRNA will be totally different.
4. **Degenerate**: Most of the amino acids have more than one codon.
    - The codon is degenerate or redundant, since there are 61 codons available to code for only 20 amino acids.
    - For instance, glycine has four codons. The codons that designate the same amino acid are called synonyms. Most of the synonyms differ only in the third (3′ end) base of the codon.
5. **Universality**: The same codons are used to code for the same amino acids in all the living organisms. Thus, the genetic code has been conserved during the course of evolution. Hence genetic code is appropriately regarded as universal.
    - There are, however, a few exceptions. For instance, AUA is the codon for methionine in mitochondria. The same codon (AUA) codes for isoleucine in cytoplasm. With some exceptions noted, the genetic code is universal.

Wobble hypothesis (put forth by Crick,)

- *Wobble hypothesis, is the phenomenon in which a single tRNA can recognize more than one codon*.
- This is due to the fact that the third base (3′-base) in the codon often fails to recognize the specific complementary base in the anticodon (5′-base).
- Wobbling is attributed to the difference in the spatial arrangement of the 5′-end of the anticodon. The possible pairing of 5′-end base of anticodon (of tRNA) with the 3′-end base of codon (mRNA) is as follows





Codon bias:

Many amino acids have multiple codons. However, the organisms prefer to use one or two codons (not all of them), and this phenomenon is referred to as codon bias. It is variable, depending the organism. As a result of codon bias, low amounts of tRNAs for the rarely used codons are made.